ZIAD MOHAMED

Third Year Computer Science Student @MSA University & University of Greenwich | Software Engineering Major

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Summary

I am a third-year Computer Science student with a strong interest in developing innovative software solutions that improve user experience and solve real-world problems. My academic background is supported by practical experience in software development, C++/C# programming, web technologies, and embedded systems.

I have completed internships where I contributed to the development of a large-scale event registration system and worked on core C++ programming tasks. My project work includes a smart pharmacy system that integrates web and embedded technologies, a Python-based resource allocation and deadlock detection tool, a full-featured art gallery e-commerce website, and a 2D game developed in C++ with AI-driven gameplay mechanics.

My technical skills include proficiency in C#/C++, Python, HTML, CSS, JavaScript, PHP (OOP and MVC), MySQL, and Arduino. I have also obtained certifications from reputable institutions such as AWS, IBM, and Google in areas like cloud computing, MongoDB, and code optimization.

I am eager to apply my skills in dynamic, fast-paced environments and to continue learning and growing as a developer. I welcome opportunities to contribute to impactful projects and collaborate with like-minded professionals.

Experience

Values Group — Cairo, Egypt

Software Development Intern – Event Registration System — 05/2025 - 05/2025

- Developed the complete registration system for the Grand Marriage Show Expo during a 3-day internship.
- Collaborated with a teammate to deliver a working solution under tight deadlines in a live event environment.

CodeAlpha — Egypt

C++ Programming Intern — 03/2025 - 03/2025

Education

Modern Sciences and Arts University

Student, Computer Science — 01/2023 - 01/2027

University of Greenwich

Student, Computer Science — 01/2023 - 01/2027

Rowad El-Mostaqbal Languages School

High School Diploma, Mathematics — 01/2014 - 01/2022

Projects

2D Platformer Game Using C# and Windows Forms — Cairo, Egypt

05/2025 - 06/2025

We've successfully designed and developed a 2D platformer game using C# and Windows Forms as part of the Multimedia (CS232) course in Spring 2025. The project was built entirely from scratch, applying core and advanced C# concepts to create a complete desktop game experience with interactive gameplay elements and visually dynamic scenes. Game Development We utilized the game logic using custom classes and lists. The design focused on user interaction, game physics, and immersive movement mechanics such as character jumps, enemy behavior, and environmental dynamics like elevators and ladders. Technologies Used: Language: C# Framework: .NET Framework (Windows Forms) Features: 1- Horizontal scrolling game environment for seamless navigation 2- Complex jump logic with increasing difficulty levels 3- Enemies with multiple behaviors to challenge the player 4- Bullet system supporting both single and multi-shot modes 5- Laser mechanics for advanced attack functionality 6- Interactive ladders and elevators for vertical movement 7- Creative level design with high attention to game flow and playability Collaboration The project was a joint effort between myself and my teammate Nada. We collaborated on both logic design and user interface to ensure a smooth and polished gameplay experience. This game not only reflects our technical growth in C# and Windows Forms but also our creativity in building engaging and functional games from the ground up.

Projects

Resource Allocation Graph – Deadlock Detection & Avoidance System

Cairo, Egypt

05/2025 - 05/2025

Language: Python, Graph Theory Developed a graphical desktop application that models the Resource Allocation Graph (RAG) used in operating systems to simulate and analyze deadlocks. The application enables users to visually create and manipulate processes and resources, and to run both deadlock detection and avoidance algorithms. Key Contributions: Deadlock Detection: Implemented cycle detection in the RAG to identify unsafe states, using standard graph traversal algorithms. Deadlock Avoidance: Integrated logic for safe state evaluation (e.g., Banker's Algorithm) to prevent system deadlock before it occurs. Graph-Driven GUI: Built an interactive graphical user interface allowing users to dynamically create and edit the graph, assign or release resources, and visualize the system state in real time. Impact: This project showcases fundamental operating systems concepts and reinforces graph theory and algorithmic problem-solving through a practical and visual tool.

Full Smart Pharmacy System for Sakr Pharmacy – Admin Portal & Online Website

Cairo, Egypt

02/2025 - 05/2025

Our team successfully designed and developed a comprehensive smart pharmacy system for Sakr Pharmacy in Sheikh Zayed. The project combined both software and embedded system components, seamlessly integrated to enhance operational efficiency, improve customer experience, and enable real-time inventory tracking and order management. Web Development We built a fully functional online website and admin portal following best practices in Object-Oriented Programming (OOP) and the Model-View-Controller (MVC) architectural pattern. This ensured clean code structure, modularity, and scalability. Technologies Used: Frontend: HTML, CSS, JavaScript, Bootstrap Backend: PHP (OOP with MVC), MySQL, phpMyAdmin Features: User-friendly customer website for browsing products, placing orders, and tracking deliveries Secure admin portal with full control over: Product management User accounts and permissions Orders and inventory Real-time system notifications Embedded Sensor System To complement the web platform, we developed a sensor-based embedded system to automate inventory monitoring. Technical Stack: Hardware: Arduino microcontroller Programming Language: C++ Communication: Telegram API This system detects when any medicine is removed from the shelf and immediately sends a real-time notification to the pharmacist via a custom-built Telegram bot, keeping the inventory status always up to date. Live Order Tracking We also implemented a live order tracking system to enhance transparency and customer trust. Once a customer places an order, they can monitor its delivery progress in real time through the website.

Smart Calculator App (C# | Windows Forms)

Cairo, Egypt

03/2025 - 03/2025

I developed a feature-rich calculator application using C# and Windows Forms that replicates the core functionalities of a standard calculator with a clean and user-friendly interface. Key Features: Intuitive UI Design: 18 functional buttons, 1 input TextBox, and 1 ListBox to display calculation history. Basic & Advanced Operations: Supports addition, subtraction, multiplication, and division with proper operator precedence. Float Number Support: Handles decimal inputs and outputs accurately. Smart Input Handling: Includes input validation to prevent malformed expressions. Error Management: Gracefully handles division by zero and invalid operations. Editing Tools: Features Clear and Delete functions for quick editing. History Tracking: Automatically stores and displays previous calculations for user reference. This project helped me strengthen my skills in C# programming and error handling, while also improving my attention to detail and user experience design.

Art Gallery Website | HTML, CSS, JavaScript, PHP | E-Commerce & Contact System

Cairo, Egypt

11/2024 - 12/2024

Developed a fully functional Art Gallery website using HTML, CSS, JavaScript, and PHP, featuring a comprehensive database system built with phpMyAdmin for user authentication (sign-up & sign-in) and painting stock management. Integrated a checkout system supporting multiple payment methods, including Apple Pay, Visa, and Cash on Delivery (COD) for a seamless transaction experience. The website also includes a contact page, where users can submit inquiries and receive automated confirmation emails via PHP Mailer. Built and tested on a WAMP server, ensuring smooth local hosting and database efficiency. Designed for usability, security, and responsiveness, providing a seamless e-commerce experience for art enthusiasts.

Data Visualization and Statistics Using Python

Cairo, Egypt

11/2024 - 12/2024

This project involved analyzing a diabetes dataset using Python to uncover patterns and insights through statistical calculations and visualizations. Key objectives included: Using Pandas, Matplotlib, Seaborn, and NumPy for data manipulation and visualization. Creating histograms, scatter plots, box plots, and pie charts to explore relationships in the dataset. Performing statistical analysis, including mean, median, mode, standard deviation, and variance for various features like glucose levels, BMI, and age. Gaining insights into how different factors impact diabetes outcomes. Used Libraries: Pandas, Matplotlib, Seaborn, NumPy

Market Management System | C++ & Data Structures | Admin & Customer Operations

Cairo, Egypt

11/2024 - 12/2024

Developed a Market Management System using C++ and advanced data structures, integrating linked lists for efficient inventory, customer, and order management. Implemented file handling for importing and exporting data from external files, ensuring persistent storage of products, customers, and transactions. The system enables customers to browse, purchase, return products, and view receipts, while admins can manage inventory, monitor orders, and track income. Designed for scalability, security, and usability, delivering a robust and optimized market management solution.

Projects

IoT-Powered Python Calculator with Arduino

Cairo, Egypt

05/2024 - 06/2024

I am pleased to present one of my recent projects — the development of a functional calculator using Python, Arduino, and IoT hardware components. This project demonstrates the integration of software and hardware to create a real-time interactive system. Technologies and Components Used: Arduino Uno Breadboard and jumper wires LCD display Matrix keypad Python (for computational logic and serial communication) Project Features: Supports basic arithmetic operations: addition, subtraction, multiplication, and division Clear/reset functionality Real-time input and output displayed on the LCD screen Seamless communication between Python and Arduino via serial interface Key Learning Outcomes: This project strengthened my knowledge in: Embedded systems and microcontroller programming Python-to-hardware communication Circuit design and hardware prototyping Real-time input handling and display updates

IOT Embedded - Bluetooth RC Using Mobile

Cairo, Egypt

10/2023 - 11/2023

The project aims to create a remote-controlled mechanism where an embedded system, equipped with Bluetooth technology, can be controlled via a mobile application. This application serves as the interface for the user to send commands to the device, which could range from simple movements like forward, backward, left, and right to more complex operations depending on the device's capabilities.

Languages

English Proficient ●●●●○

Italian Beginner ●○○○○

Arabic Native ●●●●●

Certification

AWS Educate Getting Started with Networking — Amazon Web Services (AWS)

Microsoft Excel — Coursera Project Network

AWS Node Runners for BNB Chain — Binance Academy

Code Generation and Optimization Using IBM Granite — IBM

Introduction to mongoDB — MaharaTech - ITIMooca

AWS Educate Getting Started with Compute — Amazon Web Services (AWS)

AWS Educate Getting Started with Storage — Amazon Web Services (AWS)

AWS Educate Introduction to Cloud 101 — Amazon Web Services (AWS)

Google Ads — Coursera Project Network

Investment Risk Management — Coursera Project Network

Training / Courses

Data Structure

Flutter

Awards

Third Place – Deep Minds Competition

Publications

The Bad Use Of Deepfake On Ignorant people 03/2025

Intelligent Methods, Systems, and Applications (IMSA 2025) (IEEE)

Deepfake technology has spread rapidly due to the spread of artificial intelligence, as it is an essential tool in it. The use of deepfake is not limited to honorable, entertainment, and educational purposes, but rather ignorant people have used it in a bad way, as it has become a weapon that can distort reputations, spread false information, or even manipulate public opinion. It is also not limited to the individual, but rather causes significant harm to society, as it can threaten the integrity of elections and the personal and professional lives of the victims of this exploitation. It also opens the door to deliberate smear campaigns that may be used for blackmail or to silence dissenting voices. However Victims are often ignorant of their rights, both morally and legally, in this heinous act by people who misuse deep fakes, or they do not have sufficient evidence to hold these ignorant people accountable. On the other hand, current technology is unable to distinguish between truth and falsehood due to the continuous development of this technology. Therefore, awareness must be spread about how to protect against this type of threat by implementing educational programs for all segments of society about deep fakes.